

Buoyant Force

Upward force exerted by a fluid that opposes the weight of an immersed object

- Directly upwards, opposing gravity

$$F_B = \rho_{\text{fluid}} \cdot V_{\text{object}} \cdot g$$

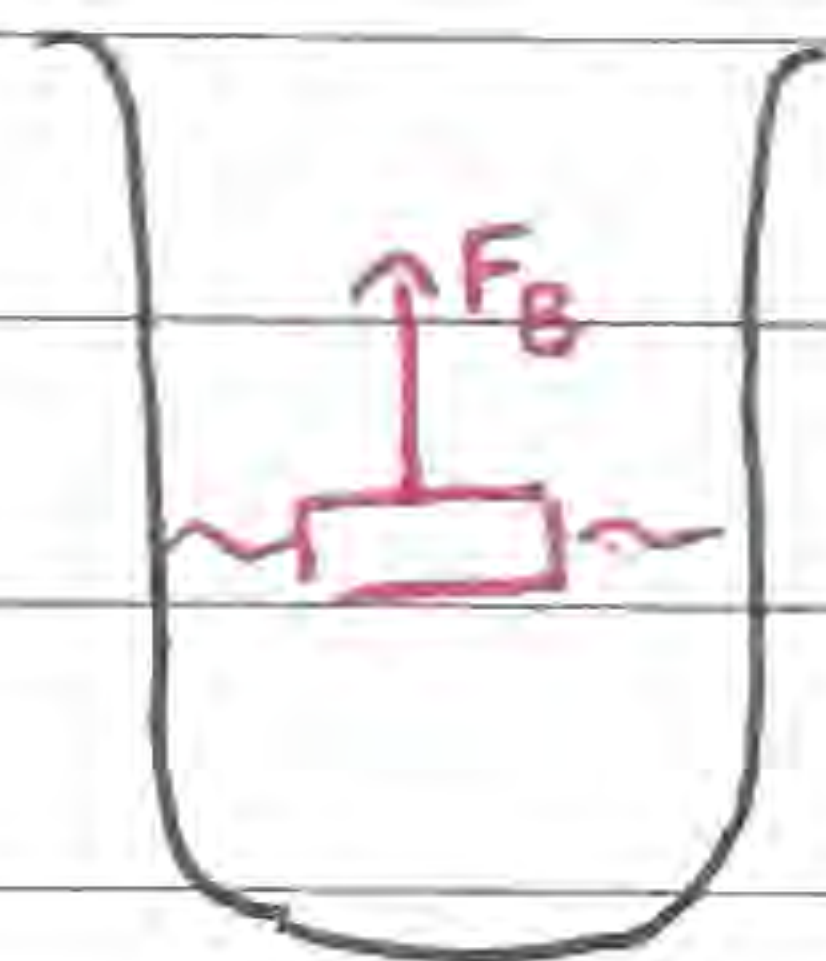
Where...

F_B - Buoyant force

ρ_{fluid} - Density of the surrounding fluid

V_{object} - Volume of the object

g - Acceleration due to the force of gravity (9.8 ms^{-2})



F_B exists in all situations when an object is submerged inside a fluid

Viscous Drag Force (F_D)

Drag force from an object's movement in a fluid

- Direction opposes object's motion

For a spherical object (at constant velocity)...

$$F_D = 6 \cdot \pi \cdot \eta \cdot r \cdot v$$

Where...

F_D - Viscous Drag Force

η - Viscosity ("eta")

r - radius of sphere

v - velocity of motion



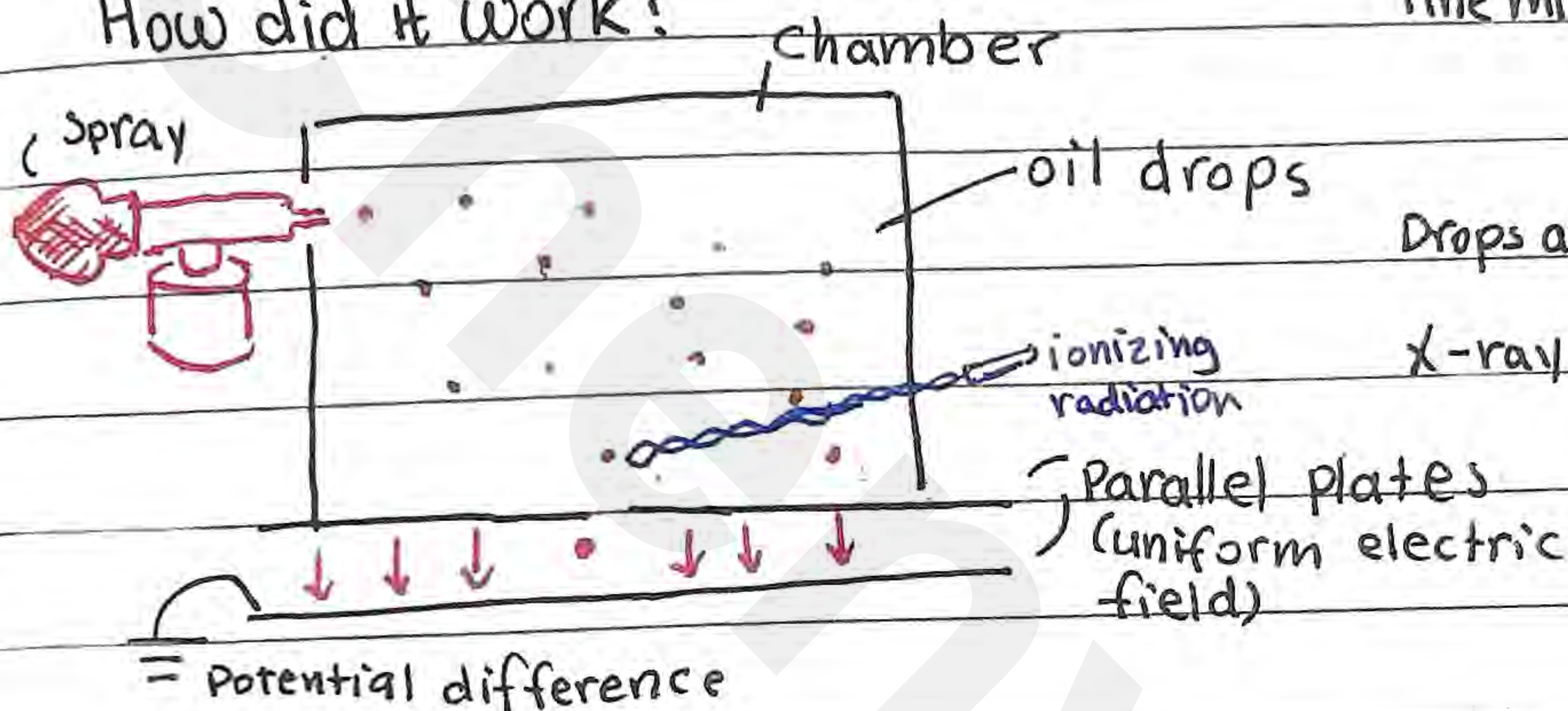
Millikan's oil drop experiment

In 1909, Millikan and Fletcher determined the value of the fundamental elementary charge

mass constant
compared to water, doesn't evaporate quickly

How did it work?

Fine mist of oil drops sprayed



Drops are ionized with

X-ray radiation

Two parts: 1. Determine radius of oil droplets (inactive battery)

falls at terminal velocity $\downarrow$ F_D $\uparrow$ F_B $\downarrow$ F_g $|v=0 \rightarrow \text{so... } F_D + F_B = F_g$

$$6\pi\eta r v + \rho_{\text{air}} V_{\text{sphere}} g = m_{\text{sphere}} \cdot g$$

knowing $m = \rho \cdot V$

> substitute $V = \frac{4}{3}\pi r^3 \rightarrow 6\pi\eta r v + \rho_{\text{air}} V_{\text{sphere}} g = \rho_{\text{oil}} \cdot V_{\text{sphere}} g$

$$r = \sqrt{\frac{9}{2} \cdot \frac{\eta \cdot v}{g(\rho_{\text{oil}} - \rho_{\text{air}})}}$$

let electric force be applied to have drop at rest

2. Determine charge of oil droplets (active battery)

at rest $\uparrow$ F_B $\uparrow$ F_E $\downarrow$ F_g $|v \neq 0$ oil drop charge typically assumed negative

> so... $F_B + F_E = F_g$

$$\rho_{\text{air}} \cdot V_{\text{sphere}} g + q \cdot E = \rho_{\text{oil}} \cdot V_{\text{sphere}} g$$

> $q = \frac{V_{\text{sphere}} g (\rho_{\text{oil}} - \rho_{\text{air}})}{E}$ $w = d \cdot f \Rightarrow V = d \cdot E; \therefore q = \frac{d}{v} \cdot g \cdot \frac{4}{3}\pi r^3 (\rho_{\text{oil}} - \rho_{\text{air}})$

Rotational Dynamics

Rigid body mechanics

• Constant angular velocity

$$\omega = \frac{\Delta \theta}{t}$$

• Constant angular acceleration

$$\omega_f = \omega_i + \alpha \cdot t$$

$$\omega_f^2 = \omega_i^2 + 2\alpha \cdot \Delta \theta$$

$$\Delta \theta = \omega_i \cdot t + \frac{1}{2} \alpha \cdot t^2$$

$$\Delta \theta = \frac{1}{2} (\omega_i + \omega_f) t$$

$$a_c = \frac{v^2}{r} = r \cdot \omega^2$$

Kinematics...

... deals with how objects move

dynamics...

... deals with why objects move

in linear dynamics, when force is applied, the object will accelerate

$$a = F_{\text{net}} \div m \quad \text{according to Newton's 2nd law}$$

in angular dynamics, when torque is applied, the object will

$$\alpha = \tau_{\text{net}} \div I \quad \text{experience angular acceleration}$$

Torque - Ability of a force to rotate a body (multiple particles)

$$\tau_{\text{net}} = I \cdot \alpha$$

where...

τ_{net} - Net torque

α - angular acceleration

I - moment of inertia of body

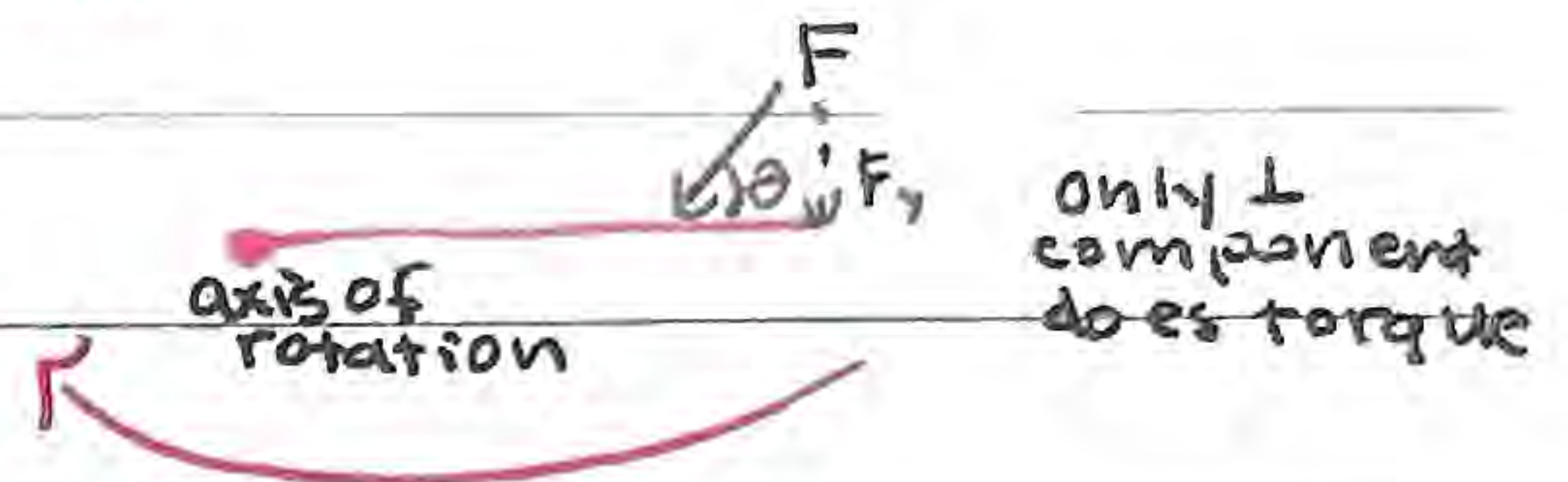
Torque (moment of a force) (N·m) ^{same but not J work!}

Torque is a vector (counterclockwise + ↺, clockwise - ↻)

Depends on how much force is applied and how far from axis of rotation it is applied on

max. torque when force is perpendicular
min. when parallel

$$|\vec{\tau}| = |\vec{r}| \cdot |\vec{F}| \cdot \sin\theta$$



where...

$\vec{\tau}$ - Torque

$\vec{r}$ - Distance of applied force from axis of rotation

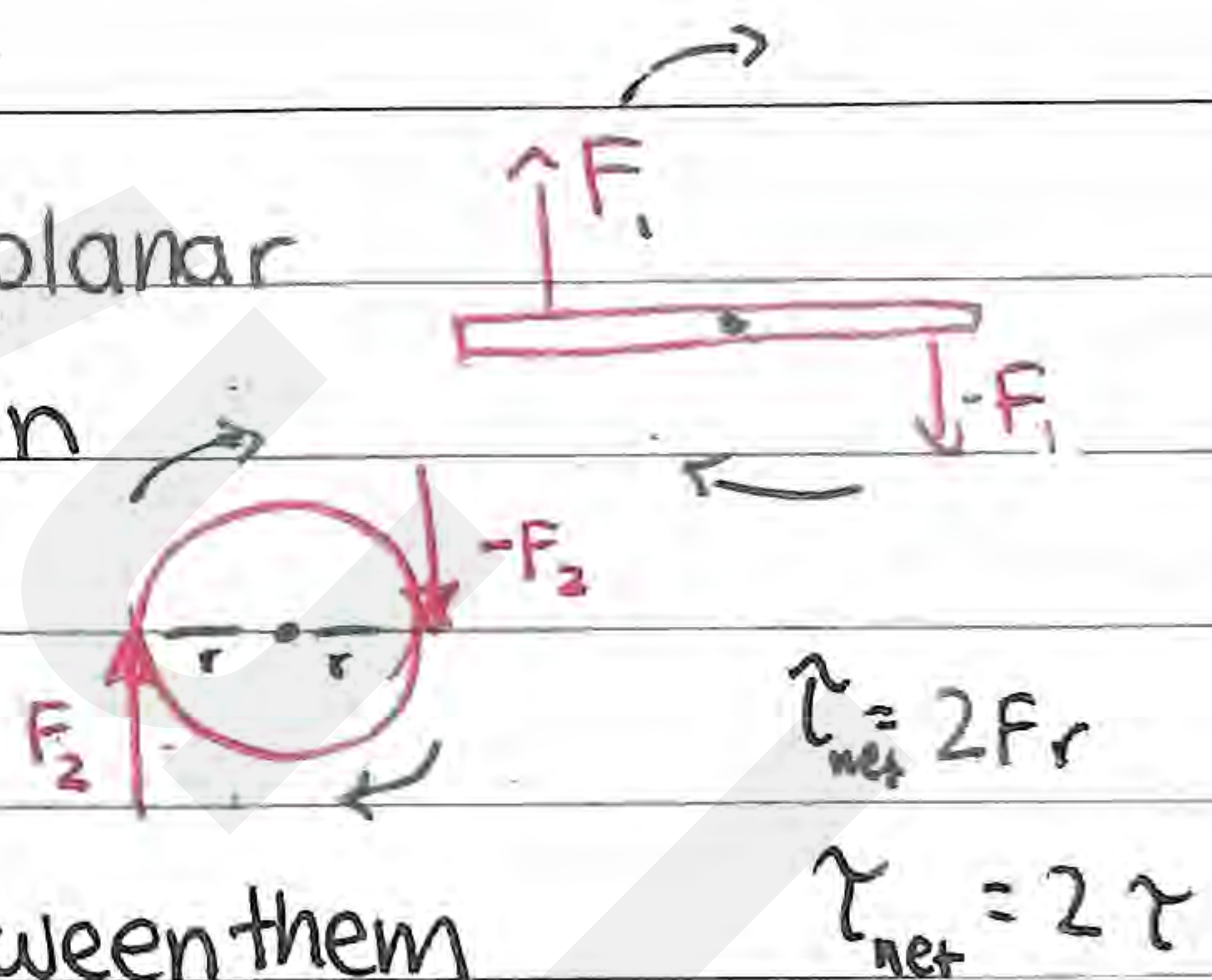
$\vec{F}$ - Applied force

θ - Angle between applied force and pivot line

Couples

Pair of equal and opposite coplanar forces to only produce rotation

- Equal in magnitude
- Opposite in direction
- Perpendicular to distance between them



Thus, no net linear force but produces torque

• Since they are identical in magnitude and distance from pivot, their torques are equal. Thus net torque is double the individual

Net torque is sum of vector torques

Rotational Equilibrium - Net torque is zero

typically used as next step to translational equilibrium

- Analogous to translational equilibrium but with torque
- Since $\alpha = 0$, you can choose any point as rotational axis (choose one that has most unknowns)

Rotational Kinetic Energy

Let's think of a body of particles



when the body is rotated, every particle follows a circular path but of different radii

The kinetic energy of an individual particle follows..

$$E_k = \frac{1}{2} m v^2$$

So the sum of the individual kinetic energy make up the total rotational kinetic energy....

$$E_k = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 + \frac{1}{2} m_3 v_3^2 + \dots$$

where... $v_i = r_i \omega$ (ω is same for all)

$$\begin{aligned}
 E_k &= \frac{1}{2} m_1 r_1^2 \omega^2 + \frac{1}{2} m_2 r_2^2 \omega^2 + \frac{1}{2} m_3 r_3^2 \omega^2 + \dots \\
 &= \frac{1}{2} (\sum m_i r_i^2) \omega^2 \\
 &= \frac{1}{2} I \omega^2
 \end{aligned}$$

I here is the moment of inertia

It measures the distribution of mass in a body

Moment of Inertia

- The IB does not require custom calculations for moment of inertia

$$I = \int R^2 dm$$

- The moment of inertia for common objects include...

1. Disc or cylinder

↑ For IB
we assume
same if it also
has a hole



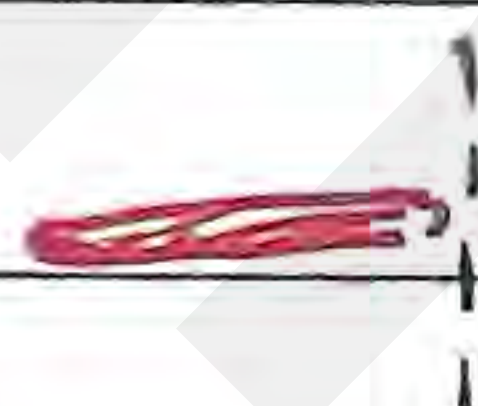
$$I = \frac{1}{2} MR^2$$

2. Sphere



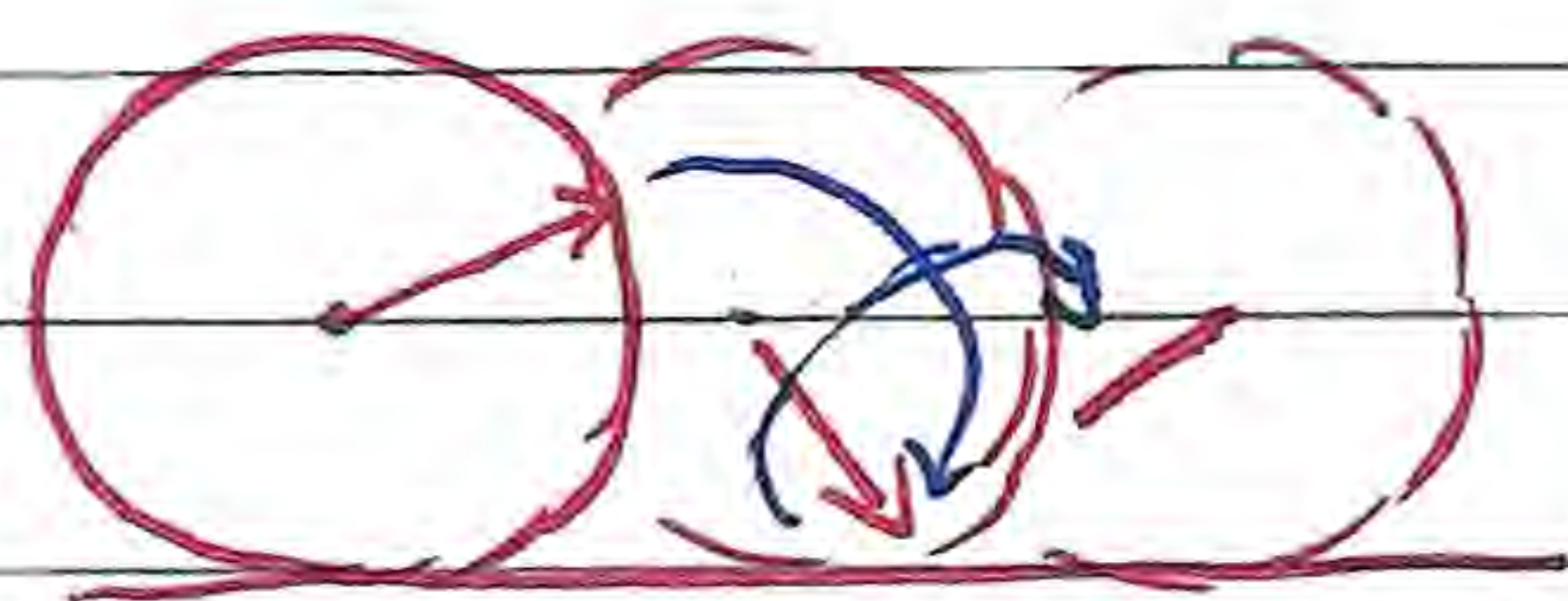
$$I = \frac{2}{5} MR^2$$

3. Rod



$$I = \frac{1}{3} ML^2$$

Rotational kinetic energy is useful for problems with an object that is rolling without slipping



The object rolls but doesn't slide

↳ i.e. not 

The force that makes this possible is frictional force (static)

↑ at the point of contact, rel. velocity is zero

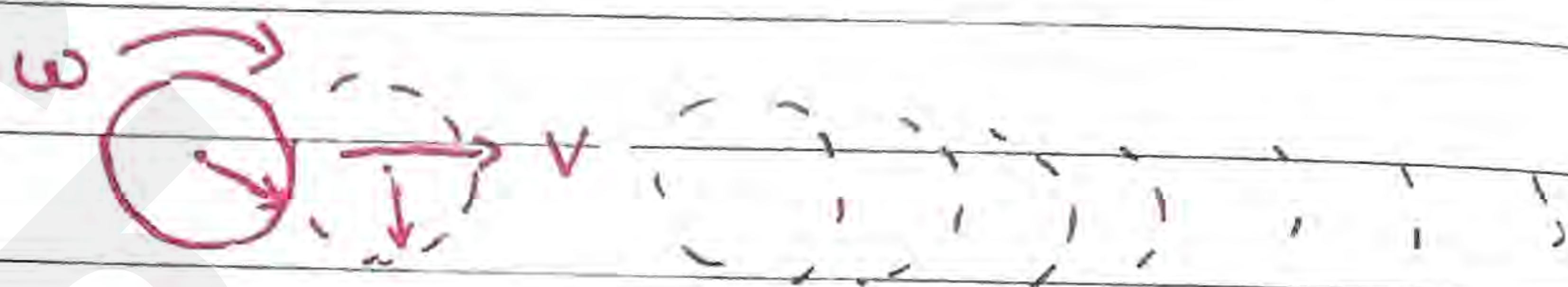
As such, the total kinetic energy sums both translational and rotational kinetic energy

$$E_k = E_{k, \text{translat.}} + E_{k, \text{rotation.}}$$

Rolling without slipping

The kinetic energy formula would be...

$$E_k = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$$



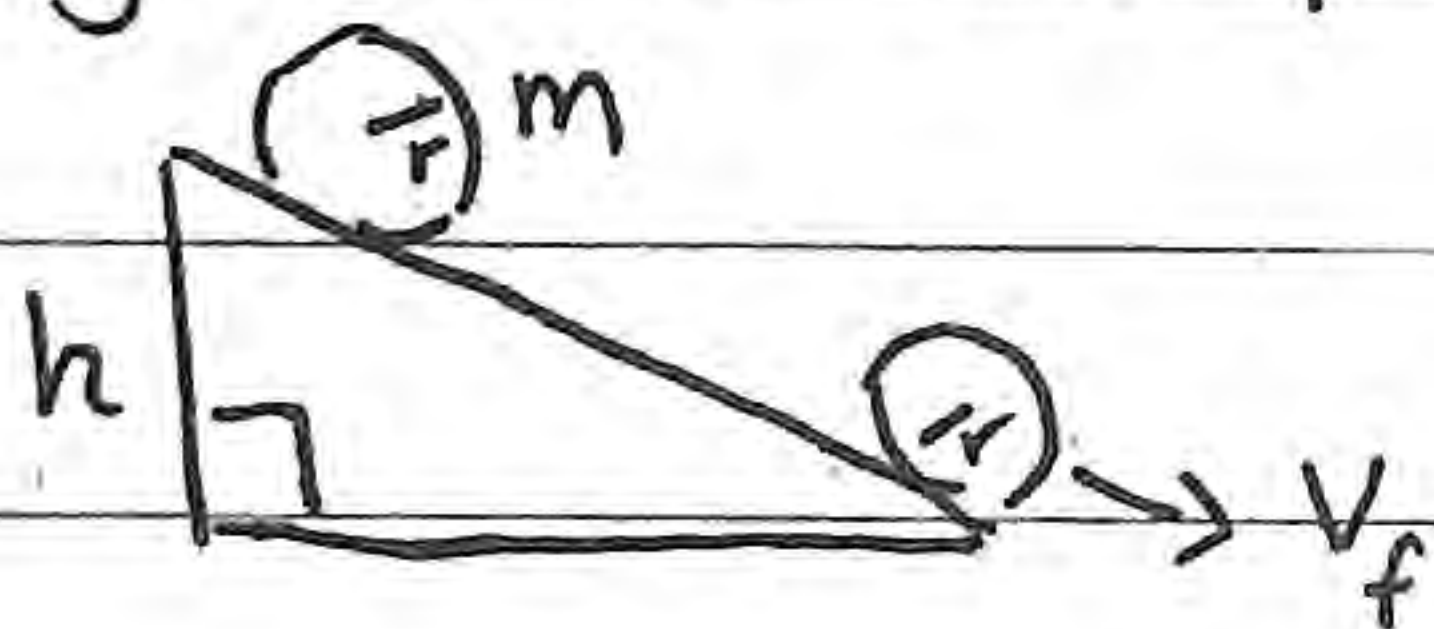
However, we can simplify this even further...

$$V_{\text{translat.}} = V_{\text{tang.}} = r \cdot \omega$$

which simplifies to...

$$E_k = \frac{1}{2} v^2 \left(m + \frac{I}{r^2} \right)$$

E.g. in a ramp, find final velocity



$$E_{p_i} = E_{k_f}$$

$$m \cdot g \cdot h = \frac{1}{2} m v_f^2 + \frac{1}{2} I \omega^2$$

$$= \frac{1}{2} m v_f^2 + \frac{1}{4} m v_f^2$$

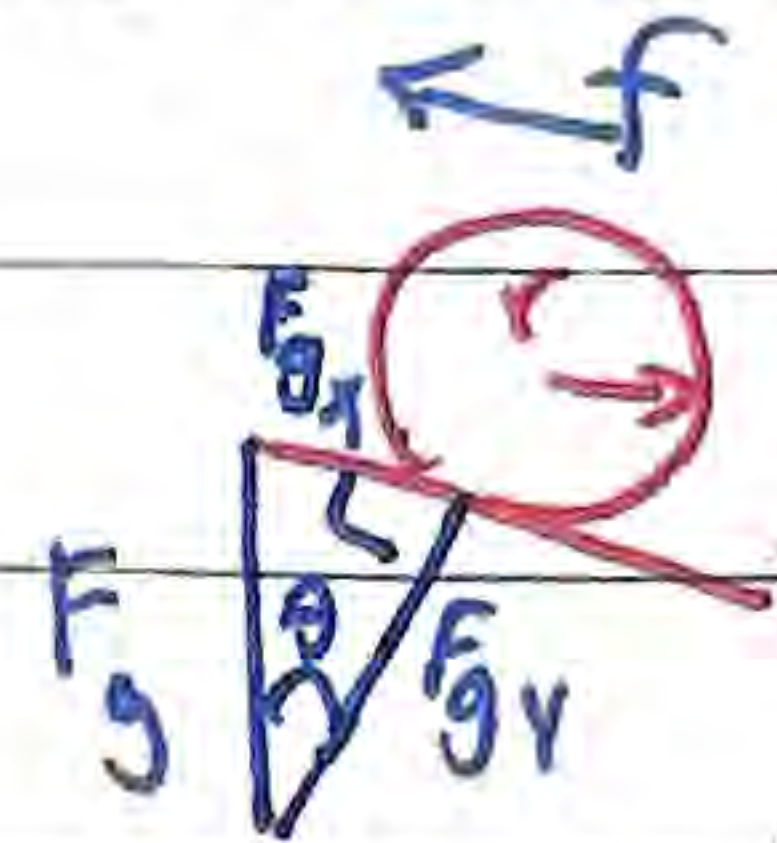
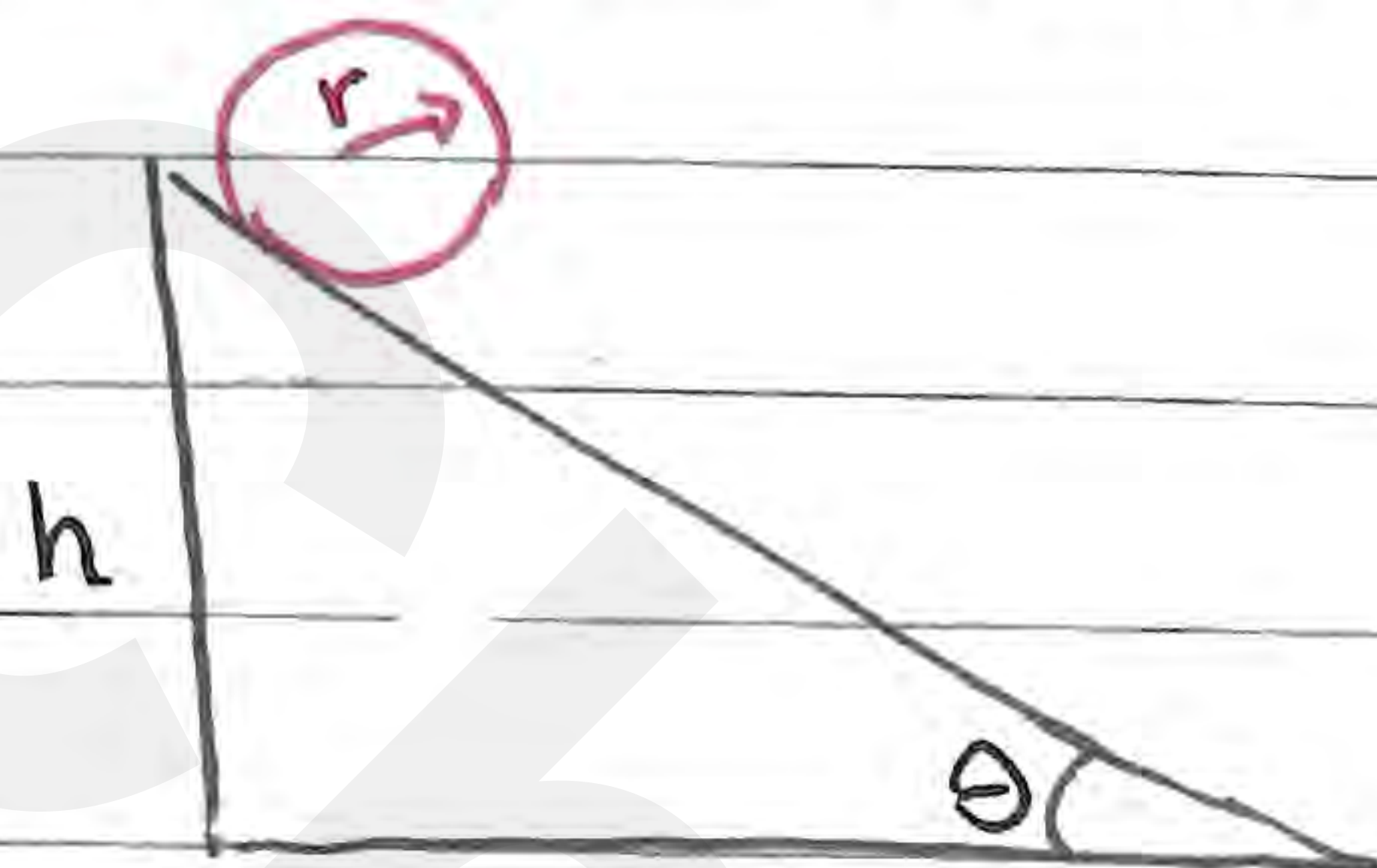
$$v_f = \sqrt{\frac{4}{3} g h}$$

$$I = \frac{1}{2} m r^2$$

Rolling without slipping

How about when asked to find acceleration?

E.g.



Force

$$m \cdot g \cdot \sin \theta - f = m \cdot a$$

$$\downarrow \quad \bar{I} = \frac{1}{2} m r^2$$

$$\tau = \bar{I} \alpha = f \cdot r \quad \text{Torque}$$

$$\therefore f \cdot r = \bar{I} \frac{a}{r} \Rightarrow f = \frac{\bar{I}}{r^2} \cdot a = \frac{1}{2} m a$$

combine

$$> m \cdot g \cdot \sin \theta - \frac{1}{2} m a = m \cdot a \Rightarrow \boxed{a = \frac{2}{3} g \cdot \sin \theta}$$

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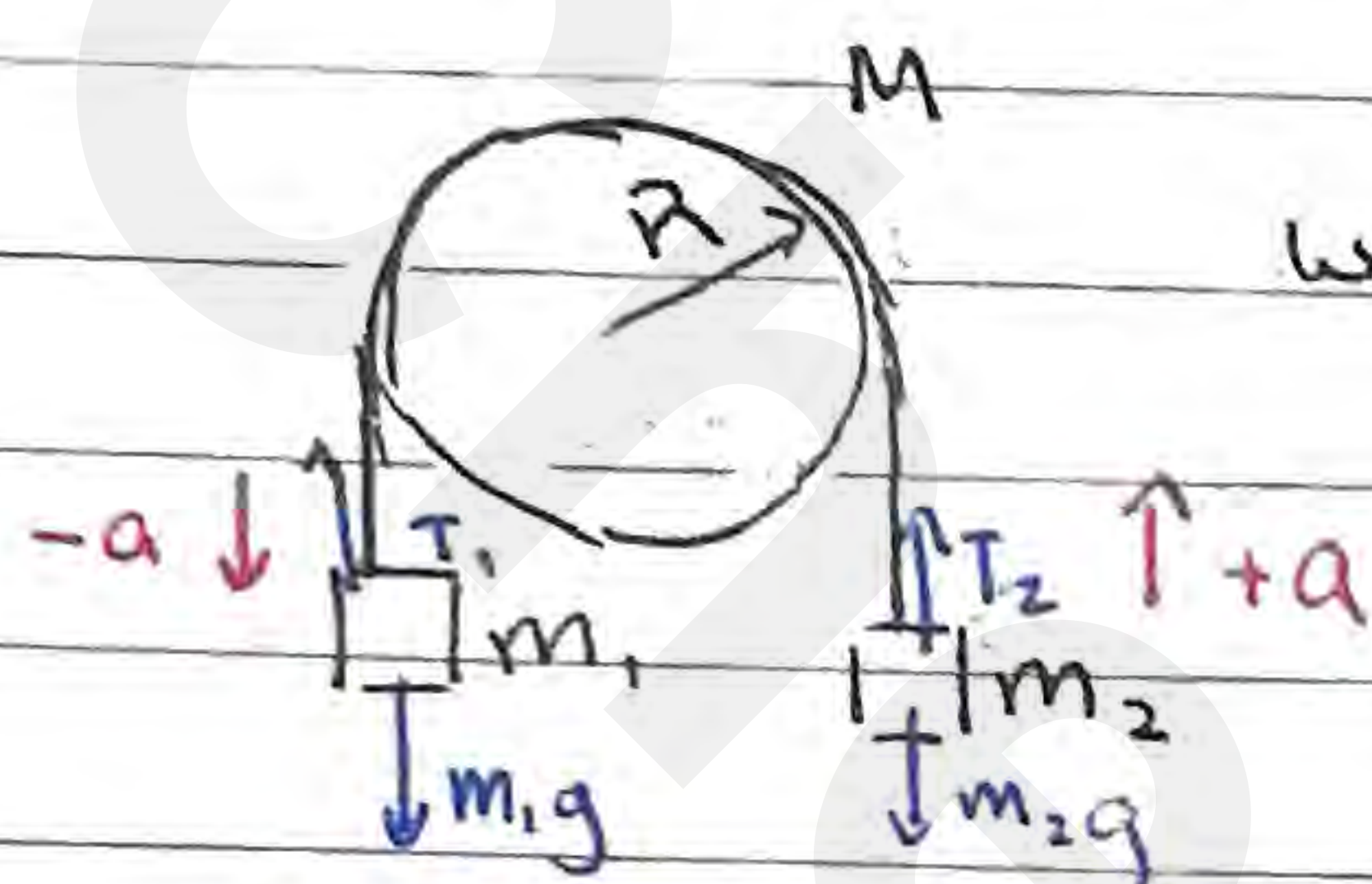
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Atwood's Machine with mass

We previously solved massless and frictionless pulley systems with tension, but what if the pulley itself has mass? (still frictionless)



where $m_1 > m_2$

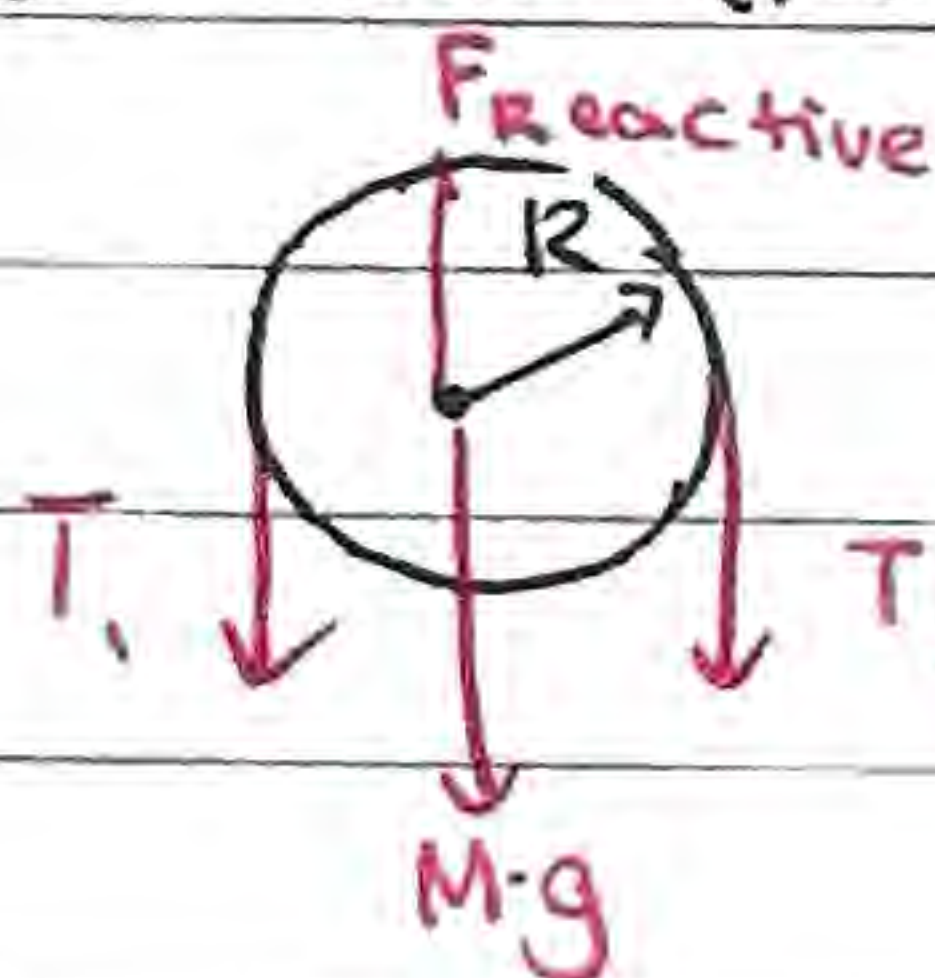
same acceleration since same rope

However, tension is different since pulley system is no longer massless

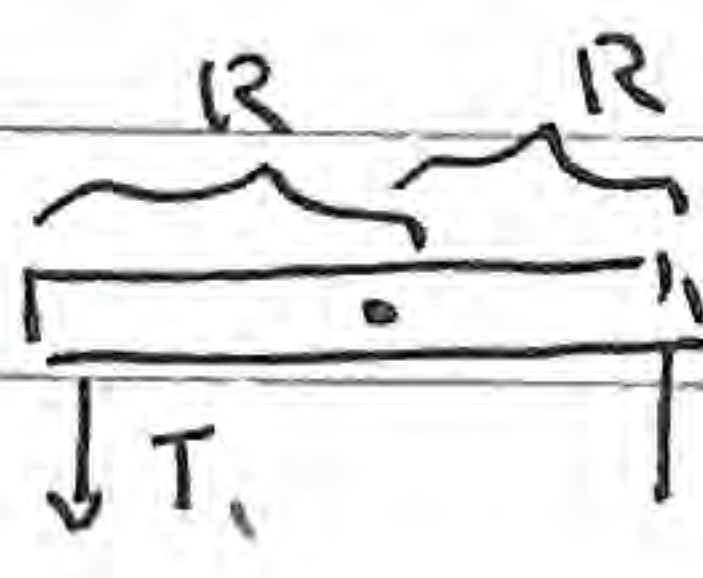
$$\textcircled{1} T_1 - m_1 g = -m_1 a$$

$$\textcircled{2} T_2 - m_2 g = m_2 a$$

We can then use torque to further analyze it



$|F_{\text{reactive}}| = |M \cdot g|$ (held to wall, so we ignore)



we can think of the rotation as

$I = \frac{1}{2} m R^2$ for cylinder

$$\tau_{\text{net}} = I \cdot \alpha = T_1 R - T_2 R = \left(\frac{1}{2} m R^2 \right) \left(\frac{a}{R} \right)$$

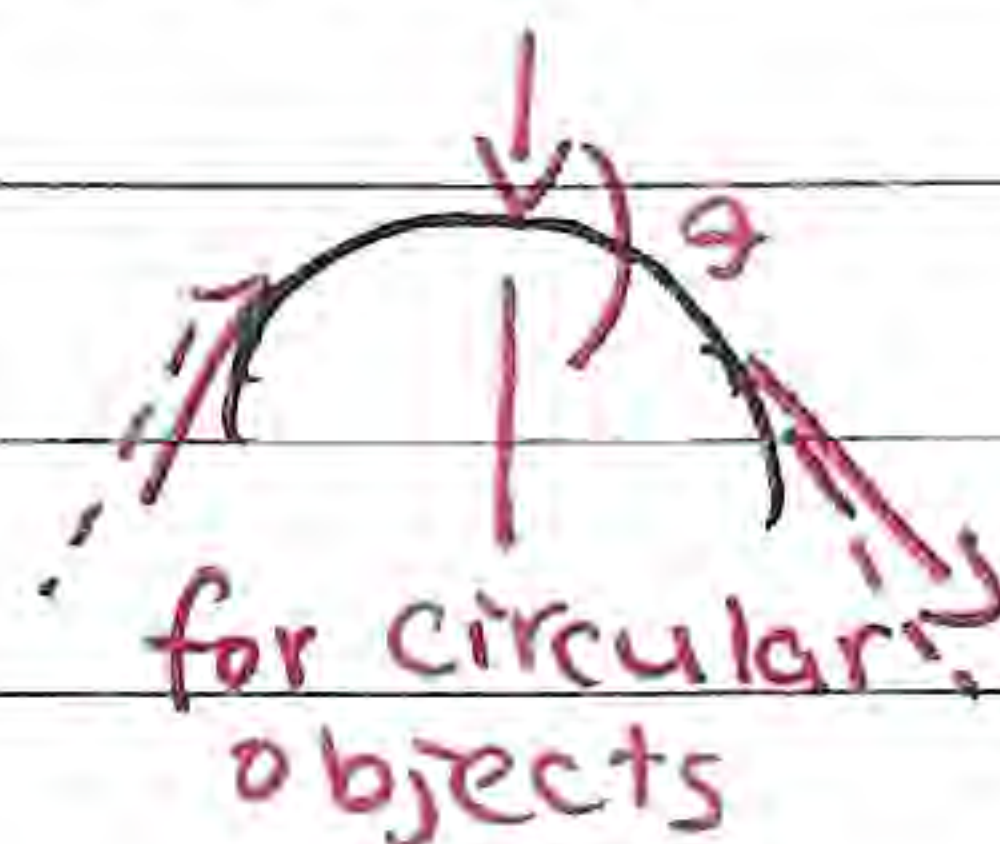
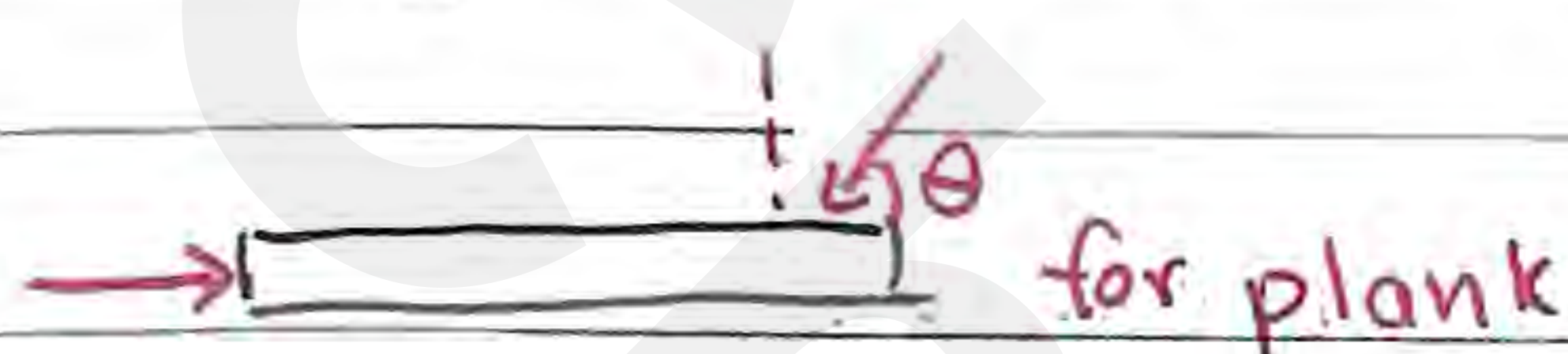
$$\textcircled{3} \therefore T_1 - T_2 = \frac{1}{2} M a$$

when $m=0$ (massless) formula matches mass system for $T_1 = T_2$

$$m_1 g - m_1 a - m_2 g - m_2 a = \frac{1}{2} M a \Rightarrow a = \frac{(m_1 - m_2) g}{m_1 + m_2 + \frac{1}{2} M}$$

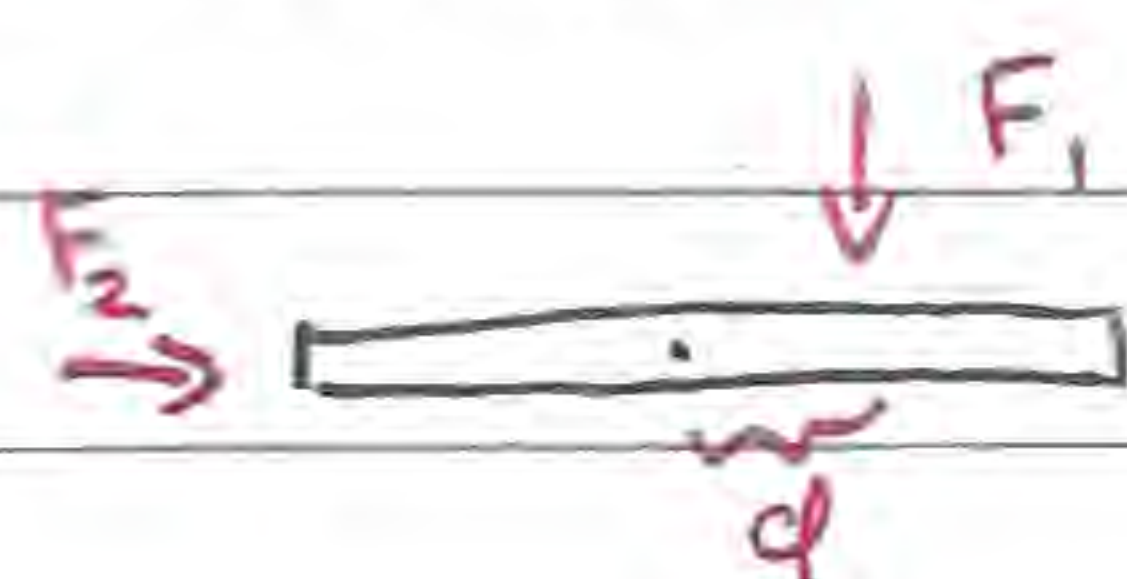
Atwood machines

* Note that for circular objects, exerting a force will result in a torque calculated from the angle between the radius and force

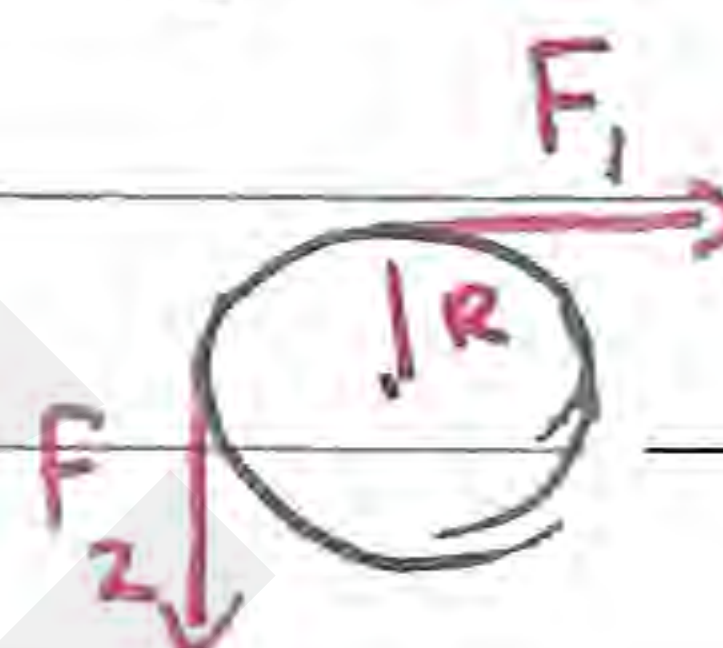


In other words, two forces

that are perpendicular on a plank cannot do the maximum torque. But, that might not be the case for circular



$$\tau_{\text{net}} = F_1 \cdot d$$



$$\tau_{\text{net}} = F_2 \cdot R - F_1 \cdot R$$

Angular Momentum

Recall linear momentum...

$$\underline{\vec{p}} = m \cdot \vec{v}$$

Angular momentum would revolve about angular velocity and moment of inertia...

$$\underline{L} = I \cdot \omega$$

It is a vectorial quantity but no need to do vector analysis for IB.

Angular Momentum

Writing Newton's second law in terms of momentum...

$$\boxed{F_{\text{net}} = \frac{\Delta p}{\Delta t}} \quad \text{and thus...} \quad \boxed{\tau_{\text{net}} = \frac{\Delta L}{\Delta t}}$$

Change in momentum is impulse...

$$\boxed{J = F_{\text{net}} \cdot \Delta t = \Delta p} \quad \text{and thus...} \quad \boxed{J = \tau_{\text{net}} \cdot \Delta t = \Delta L = \Delta(I \cdot \omega)}$$

Note that in angular, we replace mass with moment of inertia and velocity/displacement with their angular counterparts (force to torque).

Thus the following formulas are present

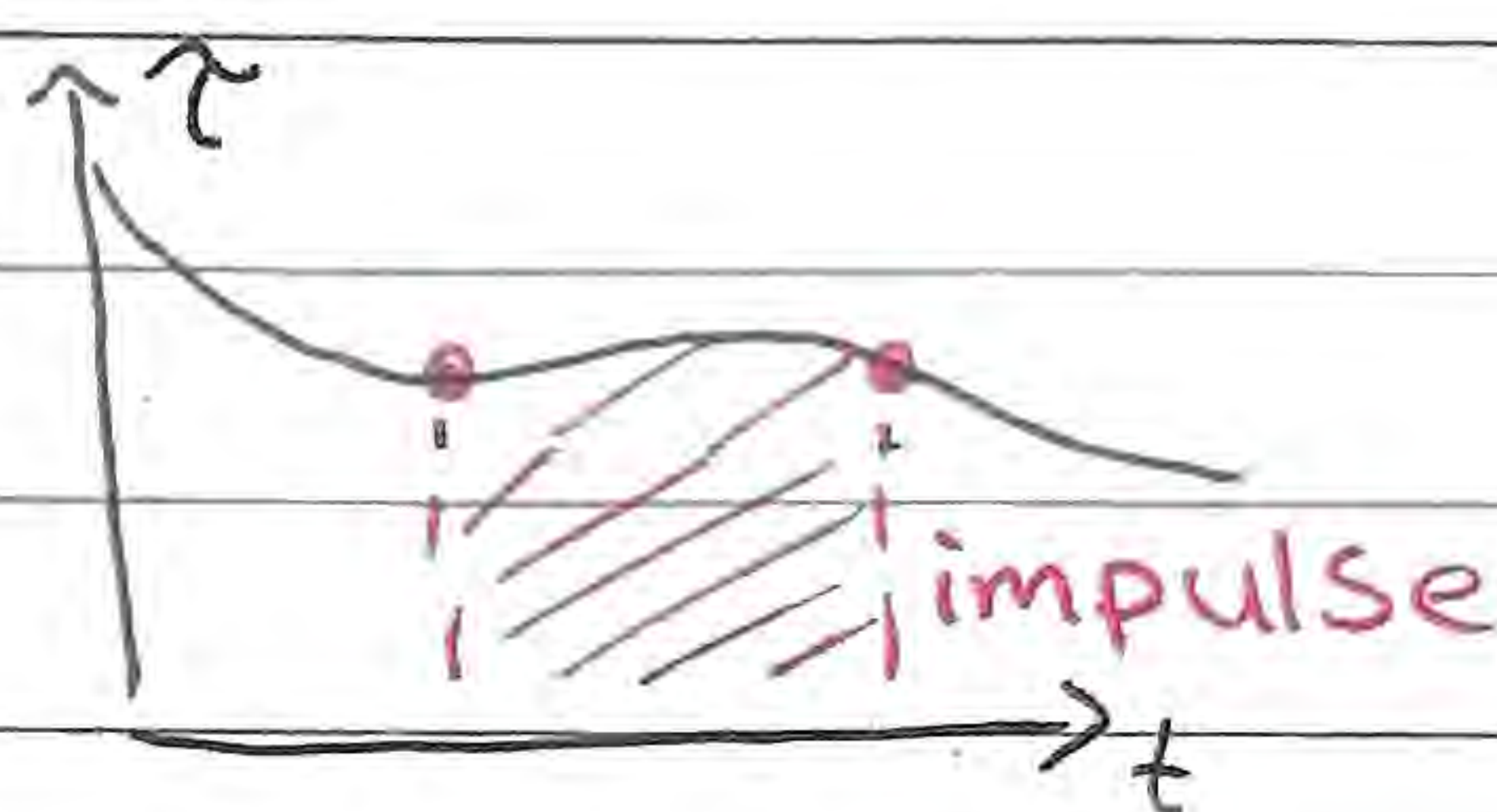
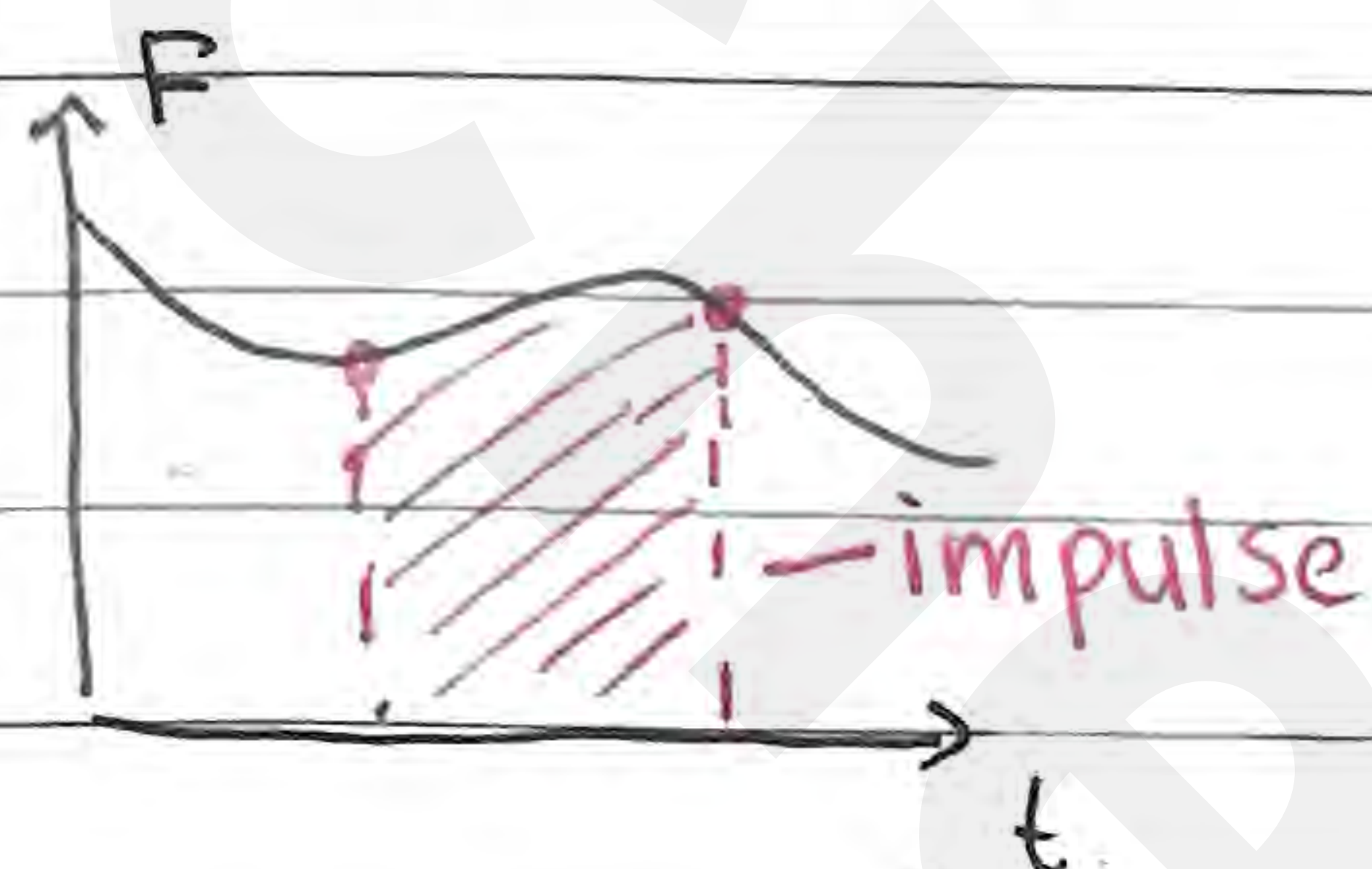
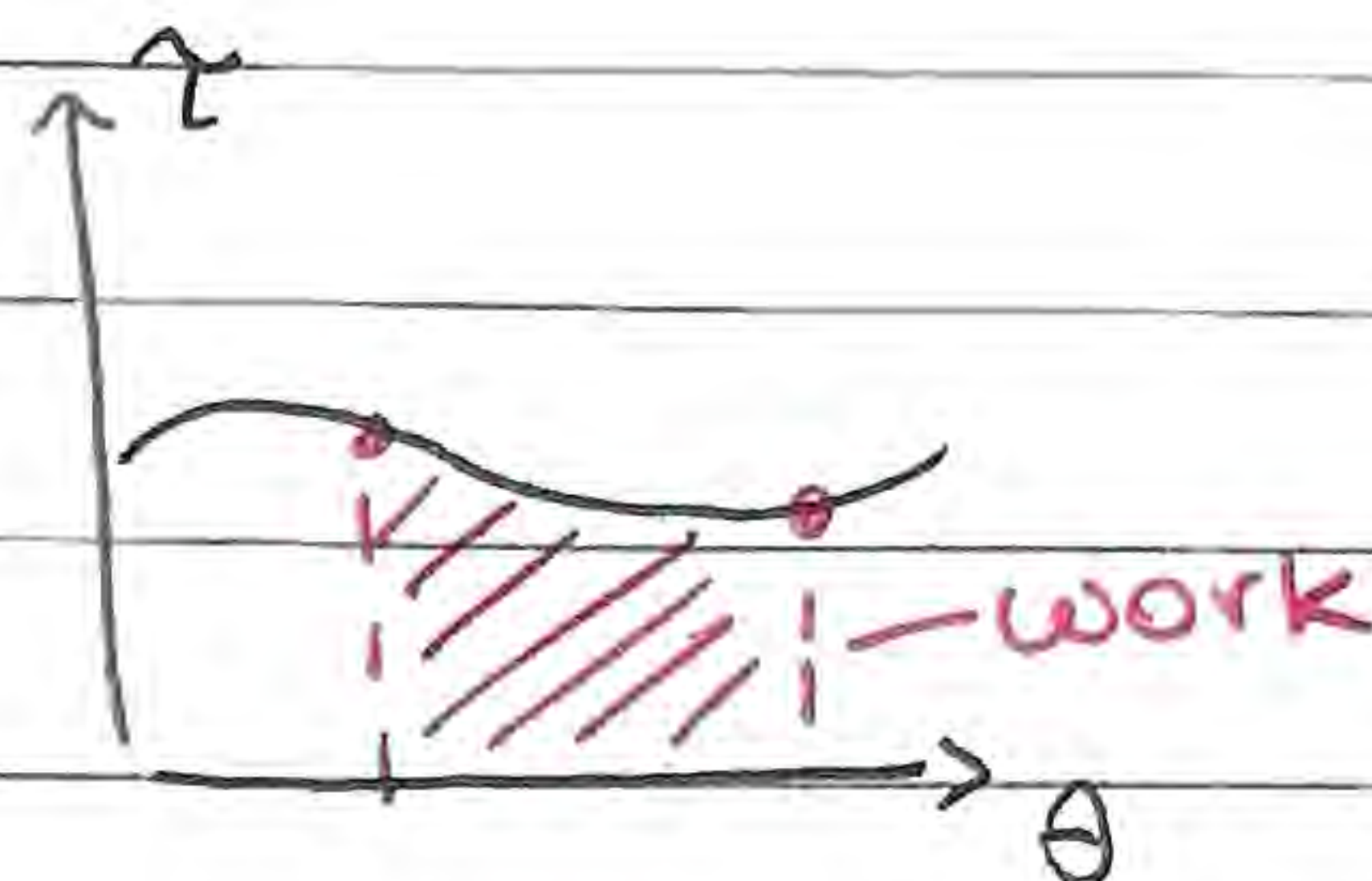
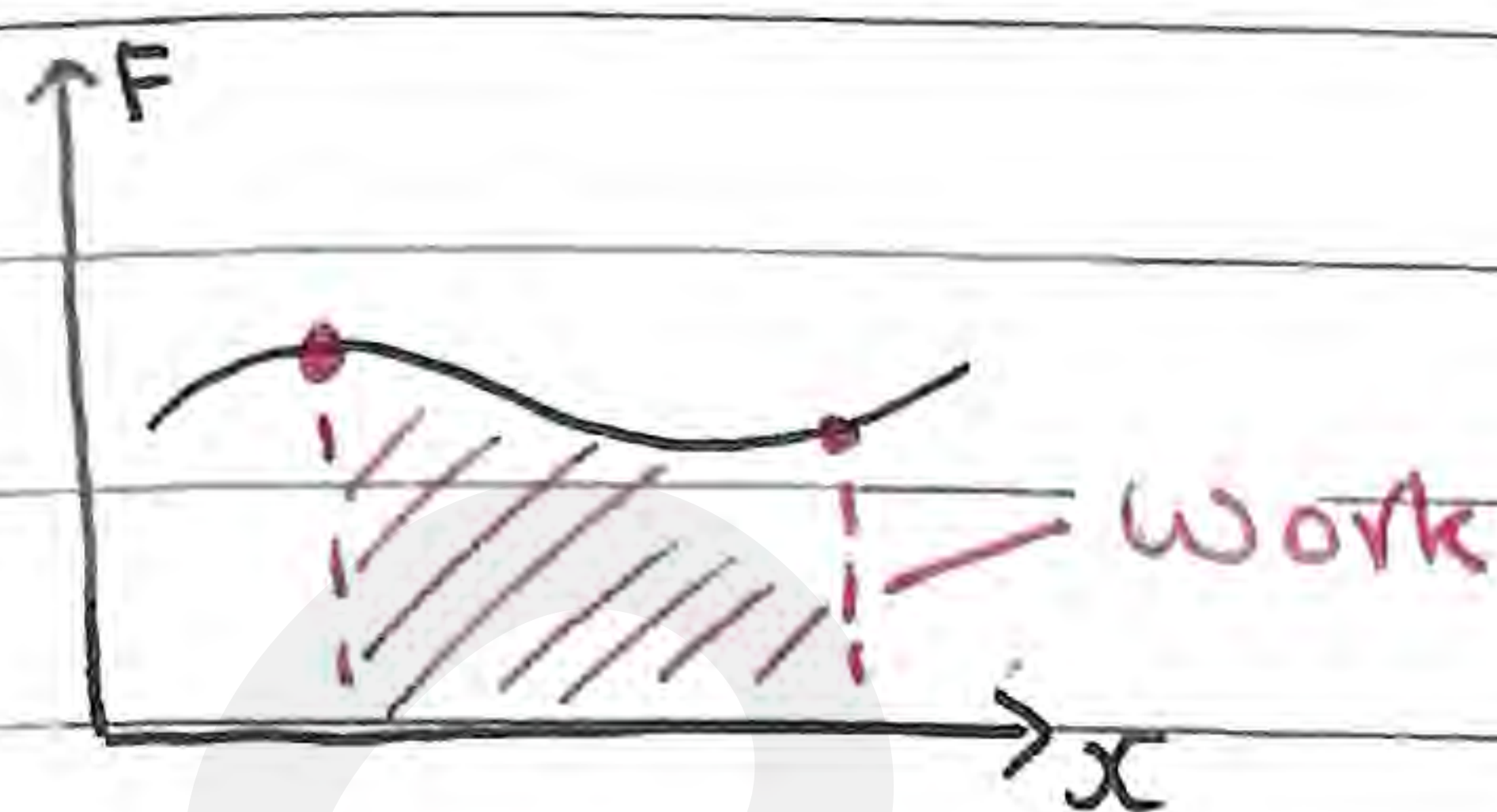
$$\begin{array}{ll} > W = \Delta E_{k, \text{transl}} & W = F \cdot s = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2 \\ & \downarrow \qquad \qquad \qquad \downarrow \end{array}$$

$$\begin{array}{ll} > W = \Delta E_{k, \text{rot}} & W = \tau \cdot \theta = \frac{1}{2} I \omega_f^2 - \frac{1}{2} I \omega_i^2 \end{array}$$

$$\begin{array}{ll} > P = \frac{W}{t} = F \cdot v & E_k = \frac{p^2}{2m} \\ & \downarrow \qquad \qquad \qquad \downarrow \end{array}$$

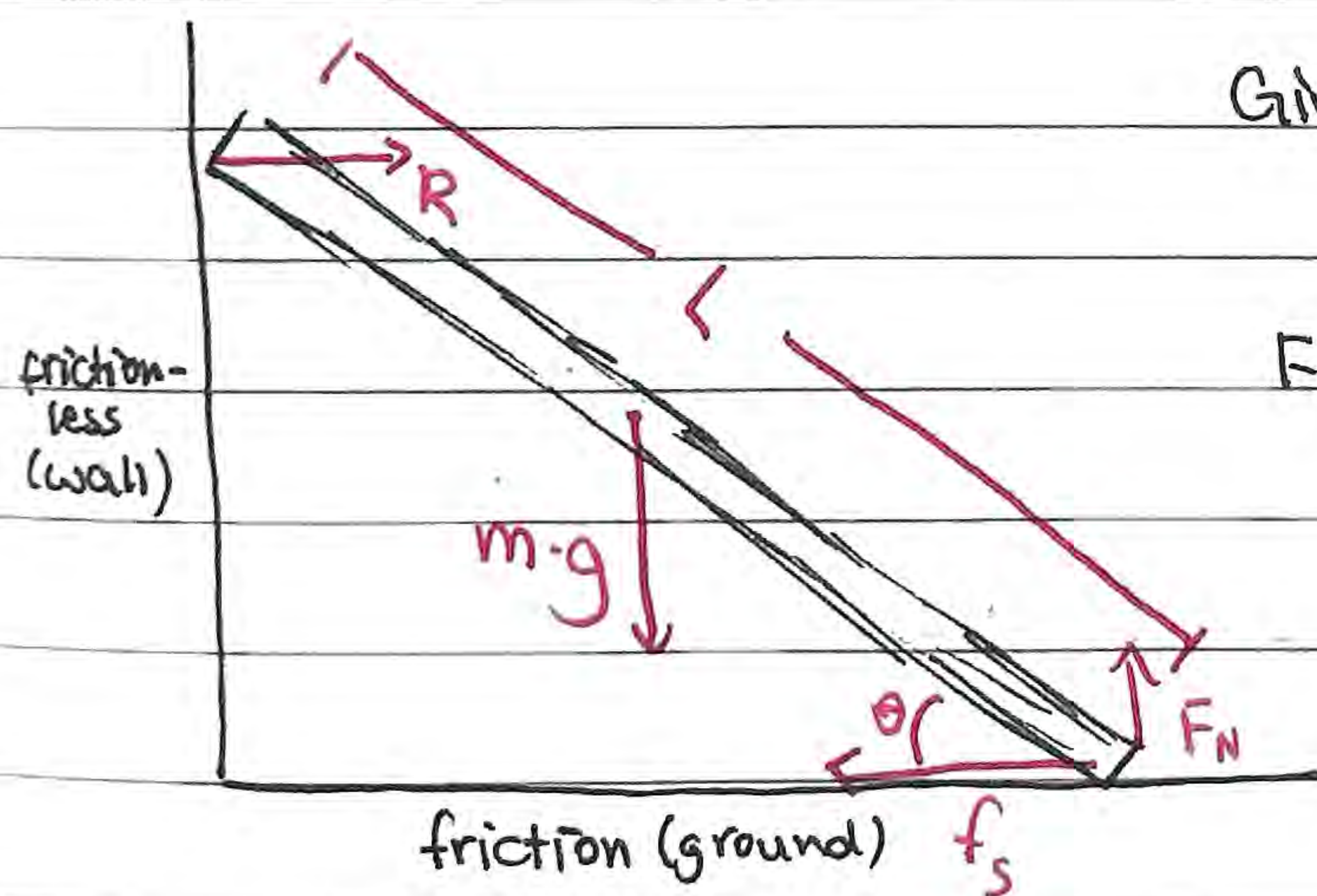
$$\begin{array}{ll} > P = \frac{W_{\text{rot}}}{t} = \tau \cdot \omega & E_k = \frac{L^2}{2I} \end{array}$$

Angular graphs



Maximum force occurs when there is a maximum torque for the same radius

E.g. Ladder Problem



Given mass of ladder, length of ladder and angle
Find μ so ladder doesn't slide

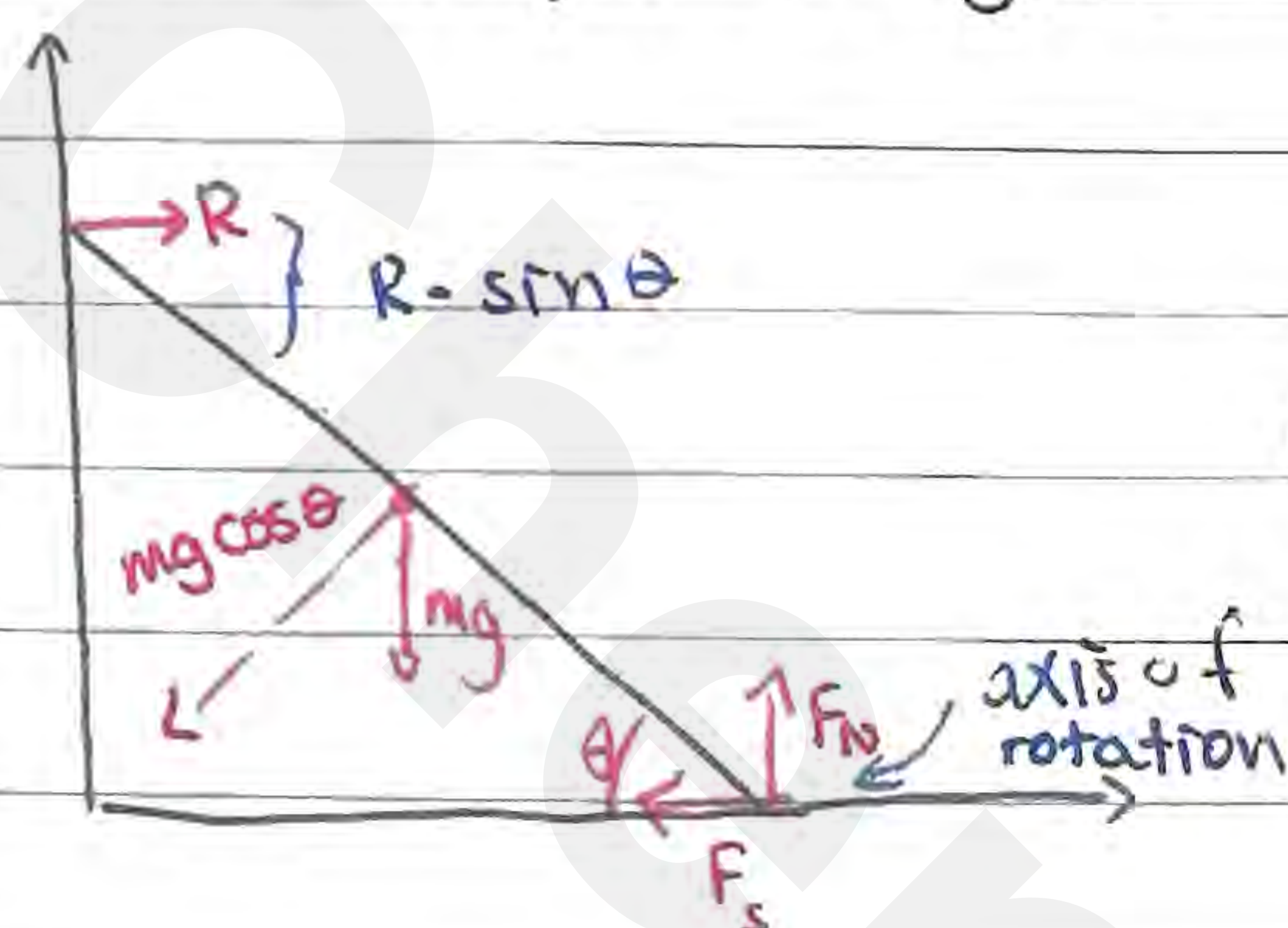
freebody diagram

Vectorial analysis: $F_{\text{net},x} = 0, F_{\text{net},y} = 0$ $R = f_{s\text{max}} = \mu \cdot M \cdot g$; $F_N = M \cdot g$

Ladder problem

$$R = \mu \cdot Mg, \quad \mu = \frac{R}{Mg}$$

With torque we get... $\tau_{\text{net}} = I\alpha = 0$



$$0 = Mg \cos \theta \frac{L}{2} - R \sin \theta L$$

$$\frac{Mg \cos \theta}{2} = R \sin \theta$$

$$R = \frac{Mg}{2 \tan \theta}$$

$$\text{Thus } \mu = \frac{1}{2 \tan \theta}$$

Conservation of angular momentum

Just like conservation of linear momentum, angular momentum is also conserved...

This happens for $\Delta L = \tau_{\text{net}} \cdot \Delta t$

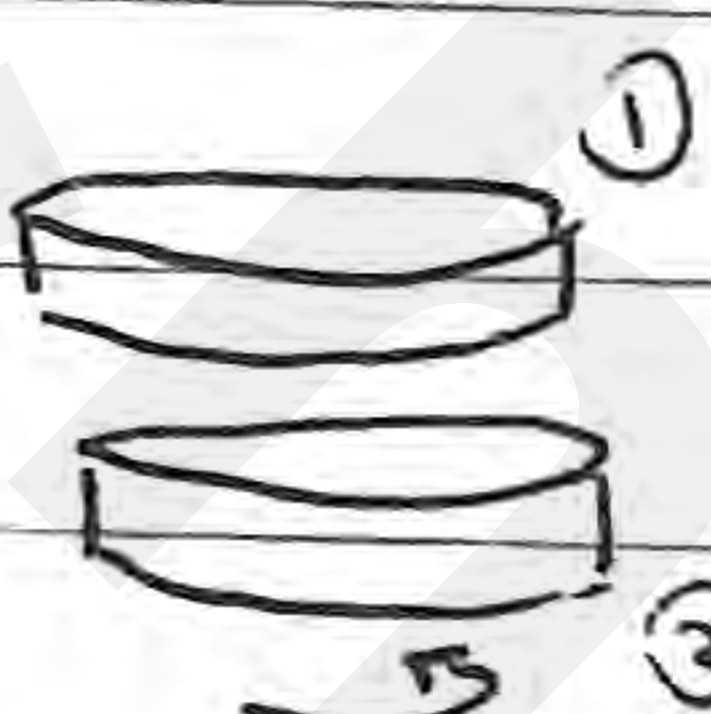
where impulse is zero, $\tau_{\text{net}} = 0$

$$\Delta L = 0 \quad \text{so} \quad \sum L_f = \sum L_i$$

Conservation of angular momentum

Also used for collisions, combining objects together. Closed, isolated systems

↳ Bringing objects to/from same point

E.g.  for two objects, what happens when they are combined?

$L_{1i} + L_{2i} = L_{1f} + L_{2f}$ when they are combined, their final rotational speed equal

$$I_1 \omega_1 + I_2 \omega_2 = (I_1 + I_2) \omega$$

$$\omega = \frac{I_1 \omega_1 + I_2 \omega_2}{I_1 + I_2}$$

Other times, it may be less obvious

E.g. explosion



There is a change in mass distribution

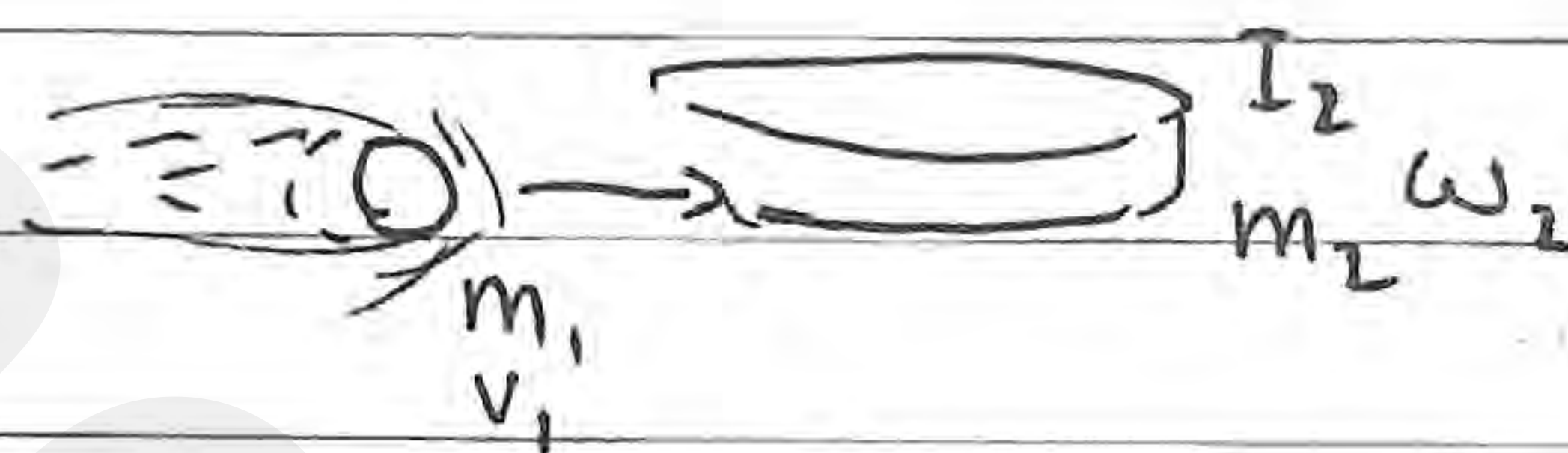
Recall $I = \sum m r^2$ so $L_i = L_f$
 $I_1 \omega_1 = I_2 \omega_2$

if $L_2 > L_1$, $r \uparrow$, $I \uparrow$ so $\omega \downarrow$ ($\omega_2 < \omega_1$)

Conservation of angular momentum

A translational movement may also have angular momentum, relative to an object in angular movement

E.g.



The angular momentum of the translational object depends on the distance to the center at which it arrives at



where $\Delta L = I \cdot \omega$

$$= I \cdot r \cdot v$$

not
radius
of spinning
object

tangential convert g
to angular

The angular
velocity
when it
arrives